

HS 123: Game Theory and Economics
BTech 5th Semester, Quiz 1
Total Marks 10, Time 55 minutes
Date: 25th Aug, 2023
[Answers should be accompanied by proper elaboration]

1. [Fiscal policy-monetary policy game] The Central Government (G) and the Reserve Bank of India (RBI) face the following hypothetical situation. The government wants to spend more money to boost the GDP. But high spending policy results in budget deficit (BD), which can lead to inflation. If G does not spend a lot, then the budget is balanced (BB). For the government BD is preferable to BB. On the other hand, the RBI can keep a high interest rate (HI) or lower the interest rate (LI). G wants the interest rate to be low, so much so that low interest in balanced budget is preferable to high interest in budget deficit. High interest policy is used by the RBI to control inflation. So, RBI prefers HI over LI in case of BD. In BB, low interest rate is preferred. For the RBI, BB with high interest rate is better than BD with high interest rate (because BB is important to it).
 - (a) Model the situation as a strategic game and find its Nash equilibrium/equilibria. Is the game similar to the Prisoner's Dilemma game? [3+1]
 - (b) Are there any dominated actions in this game? If yes, mention them. [1]
2. A mother with three children (Alia, Bikki, Chandu) discovers a broken flower pot in her house. She is more interested to find out who is the culprit, rather than punish the culprit (actually Alia broke it, the mother does not know). She asks the children to play the following game. Each child will write her/his name on a piece of paper and write "Yes, I broke it" or "No, I didn't break it" and give the paper to her. If at least one child confesses, the confessor(s) will get the normal pocket-money of Rs 20, the rest will get Rs. 50. If no one claims that she/he has broken it, then each child receives Rs. 0. Model the situation as a strategic game and write down the payoff table. Find all the Nash equilibria. Which is the focal point according to you? [2+2+1]

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 BTech, 5th Semester, Mid-Semester Test
 Total Marks: 40
 Date: 23 September, 2023

[Explain the answers in words. Without proper elaboration of answers, marks would be deducted.]

1. Players 1 and 2 are engaged in a game, each can choose x, y, z. Payoffs of player 1 for different combination of choices is given in the table below. Player 2 gets the negative of these numbers. Find the minimax solution of the game. [6]

		2		
		x	y	z
1	x	4	4	3
	y	5	1	1
	z	7	0	2

2. Find all the Nash equilibria of the game given below. Which of the equilibria, if any, correspond to a steady state if the game models pairwise interactions between the members of a single population (and why)? [3+3]

		2		
		a	b	c
1	a	3,3	3,4	0,3
	b	4,3	1,1	0,1
	c	3,0	1,0	0,0

3. Give an example of a two-player strategic game in which each player has finitely many actions and in the only Nash equilibrium both players' actions are weakly dominated. [6]
4. Consider a Cournot duopoly where the demand function is given as, $P = 50 - \frac{1}{2}Q$. The cost function of each firm is, $2q_i^2$.
- How many Nash equilibria are there and how much output do the firms produce at the equilibrium/equilibria? [3]
 - What is equilibrium price in the market, and what are the profits of firms? [3]
 - Suppose firm 2 changes its technique of production and its new cost function is, $2q_2$. What are the new equilibrium output levels? Is the new equilibrium price higher or lower compared to your answer in (b)? [4+2]

5. [Variation of Bertrand model] Suppose goods 1 and 2 are produced by firms 1 and 2. Firms choose their prices p_1 and p_2 . Each has a constant per unit c . The direct demand functions are, $D_1(p_1, p_2) = \alpha_1 - \beta_1 p_1 + \gamma_1 p_2$, and $D_2(p_1, p_2) = \alpha_2 - \beta_2 p_2 + \gamma_2 p_1$. Here $\alpha_1, \beta_1, \gamma_1, \alpha_2, \beta_2, \gamma_2 > 0$.

- (a) Find the profit functions of the firms and derive their best response functions. [4]
- (b) As p_j rises, does the optimal p_i rise or fall? Why? [3]
- (c) What is the equilibrium/equilibria set of prices? [3]

HS 123: Game Theory and Economics
BTech 5th Semester, Quiz 2
Total Marks 10, Time 55 minutes
Date: 17 Oct, 2023
[Answers should be accompanied by proper elaboration]

1. Consider the variant of Hotelling's model in which the candidates (like the citizens) care about the winner's position, and not at all about winning per se. There are two candidates. Each candidate has a favourite position; her dislike for other positions increases with their distance from her favourite position. Assume that the favourite position of one candidate is less than m and the favourite position of the other candidate is greater than m . Assume also that if the candidates tie when they take the positions x_1 and x_2 then the outcome is the compromise policy $\frac{x_1+x_2}{2}$. Find the set of Nash equilibria of the strategic game that models this situation. [6]

[Hint: First consider pairs (x_1, x_2) of positions for which either $x_1 < m$ and $x_2 < m$, or $x_1 > m$ and $x_2 > m$. Next consider pairs (x_1, x_2) for which either $x_1 < m < x_2$, or the other way; then those for which $x_1 = m$ and $x_2 \neq m$, or $x_1 \neq m$ and $x_2 = m$. Finally consider the pair (m, m) .]

2. Show that in a first-price sealed-bid auction any action profile satisfying the three conditions: the two highest bids are the same, one of these bids is submitted by player 1, and the highest bid is at least v_2 and at most v_1 , is a Nash equilibrium. [4]

HS 123: Game Theory and Economics
BTech, 5th Semester, End-Semester Test
Total Marks: 40
Date: 24 November, 2023

1. Chris Evert and Martina Navratilova are playing a game of lawn tennis. Evert can hit two kinds of shots, Down the Line (DL, a hard, straight shot), or Cross Court (CC, a soft, diagonal shot). Navratilova can prepare herself for each of them. The payoffs are given by the following table (the numbers are the percentage of times a player wins a point).

		Navratilova	
		DL	CC
Evert	DL	50, 50	80, 20
	CC	90, 10	20, 80

- (a) What are the pure strategy and mixed strategy equilibria of the game? What is the expected payoff of Evert in the mixed strategy equilibrium? [4+1]
 (b) Evert injures herself and her DL shots become weaker. As a result the payoffs change and are given by the following table. Find the mixed strategy equilibria now. Is Evert playing DL with lesser probability compared to (a)? What is her expected payoff in the equilibrium? [3+1+1]

		Navratilova	
		DL	CC
Evert	DL	30, 70	60, 40
	CC	90, 10	20, 80

2. For the following game with vNM preferences show that mixed strategies $\alpha_1 = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right)$, $\alpha_2 = \left(\frac{1}{3}, 0, \frac{2}{3}\right)$ constitute a mixed strategy Nash equilibrium. [6]

		2		
		x	y	z
1	a	6, 5	2, 4	9, 4
	b	3, 6	10, 10	10.5, 0
	c	16, 2	1, 0	4, 10

3. Arman and Bikas take turns in removing matchsticks from a pile. They start with 21 matchsticks, and Arman goes first. On each turn, each player may remove one, two, three or four matchsticks. The player to remove the last matchstick wins the game.
- (a) Suppose there are only six matchsticks left and it is Bikas's turn. What move should Bikas make to guarantee himself victory? Explain your reasoning. [2]
 (b) Suppose there are 12 matchsticks left and it is Bikas's turn. What move should Bikas make to guarantee himself victory? [2]
 (c) Now start from the beginning of the game. If both players play optimally who will win? Is there a first mover advantage? [2]
 (d) What are the optimal strategies for each player? [2]
4. Two people select a policy that affects them both by alternatively vetoing policies until only one remains (the word *veto* means reject). First, person 1 vetoes a policy, if more than one policy

remains person 2 then vetoes a policy. If more than one policy still remains, person 1 then vetoes another policy. The process continues until a policy remains unvetoes. Suppose there are three possible policies, X, Y and Z, person 1 prefers X to Y to Z, and person 2 prefers Z to Y to X.

(a) Model this situation as an extensive game and find its Nash equilibria. [2+4]

(b) Does the game have any Nash equilibrium which is not a subgame perfect equilibrium? Elaborate. [4]

5. Consider a Stackelberg duopoly where the inverse demand and cost functions are given by $P = \alpha - Q$ and $C_i(q_i) = q_i^2$ respectively ($\alpha > 0$).

(a) Find the subgame perfect output levels of the firms. [4]

(b) For a similar specification Cournot duopoly Nash equilibrium is $q_i^* = \frac{\alpha}{5}$. Compare your result in part (a) with the Cournot duopoly Nash equilibrium and comment. [2]